Deccan Education Society's

Kirti M. Doongursee College of Arts, Science and Commerce [NAAC Accredited: "A Grade"]

T.Y.B.Sc. [Computer Science]

Practical Journal

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Seat Number [681373]

Roll No:-34

Department of Computer Science and Information Technology

Department of Computer Science and Information Technology

Deccan Education Society's

Kirti M. Doongursee College of Arts, Science and Commerce[NAAC Accredited: "A Grade"]

C E R T I F I C A T E

This is to certify that Ms. **Neha Ganesh Patil** of T.Y.B.Sc. (Computer Science) with seat no. 681373 & roll no. **34** has completed 10 practicals of Paper- USCSP605 under my supervision in this College during the year 2021-2022.

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Date: / /2021

Examined by: **Remarks:**

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1.a. 2D Linear Convolution

Code

```
clc;
x =[1 ,2 ,3;4 ,5 ,6;7 ,8 ,9];
h = [1 ,1;1 ,1;1 ,1];
y = conv2(x,h);
disp(y,'Linear 2D convolution y=');
```

Output

```
"Linear 2D convolution y="

   1.    3.    5.    3.

   5.    12.   16.   9.

   12.   27.   33.   18.

   11.   24.   28.   15.

   7.    15.   17.   9.
```

1.b. Circular Convolution between two 2D matrices

Code

```
clc ;
x = [1,2;3,4];
h = [5,6;7,8];
X = fft2(x);
H = fft2(h);
Y = X.*H;
y = ifft(Y);
disp (y, 'Circular Convolution Result y = ');
```

Output

```
"Circular Convolution Result y = "

70.   68.

62.   60.
```

2 Circular Convolution expressed as linear convolution plus alias

Code

```
clc ;
x = [1 ,2;3 ,4];
h = [5 ,6;7 ,8];
y = conv2 (x,h);
y1 = [y(:,1)+y(:,$),y(:,2)];
y2 = [y1(1,:)+y1($,:);y1(2,:)];
disp(y,'Linear Convolution Result y =');
disp(y2,'Circular Convolution Expressed as Linear Convolution plus alias = ');
```

Output

```
"Linear Convolution Result y ="

  5.   16.   12.

 22.   60.   40.

 21.   52.   32.

"Circular Convolution Expressed as Linear Convolution plus alias = "

 70.   68.

 62.   60.
```

3. a. Linear Cross correlation of a 2D matrix

Code

```
clc ;
x = [3 ,1;2 ,4];
h1 = [1 ,5;2 ,3];
h2 = h1 (:,$ : -1:1);
h = h2($: -1:1 ,:);
y = conv2 (x,h);
disp(y, "Linear Cross Correlation of a 3D matrix y = ");
```

Output

```
"Linear Cross Correlation of a 3D matrix y = "

   9.   9.   2.

  21.  24.   9.

  10.  22.   4.
```

3. b. Circular correlation between two signals

Code

```
clc ;
x = [1 ,5;2 ,4];
h = [3 ,2;4 ,1];
h = h(:,$ : -1:1) ;
h = h($: -1:1 ,:);
X = fft2 (x);
H = fft2 (h);
Y = X.*H;
y = ifft (Y);
disp(y, "Circular Correlation Result y = ");
```

Output

```
"Circular Correlation Result y = "

  37.  23.

  35.  25.
```

3. c. Linear auto correlation of a 2D matrix

Code

```
clc ;
x1 = [1 ,1;1 ,1];
x2 = x1 (:,$ : -1:1);
x2 = x2($: -1:1 ,:);
x = conv2 (x1 ,x2);
disp(x, "Linear auto Correlation Result x = ");
```

Output

```
"Linear auto Correlation Result x = "

  1.   2.   1.

  2.   4.   2.

  1.   2.   1.
```

4 DFT of 4x4 gray scale image

Code

```
clc;
f=[ 1 1 1 1;1 1 1 1;1 1 1 1;1 1 1 1];
t=fft2(f);
disp(t,'display')
```

Output

```
 "display"

 16.   0.   0.   0.

  0.   0.   0.   0.

  0.   0.   0.   0.

  0.   0.   0.   0.
```

5. a. Compute discrete cosine transform

Code

```
clc;
N=[1,2,3,4]
x=dct(N);
disp(x,'DCt matrix of order four')
```

Output

```
"DCt matrix of order four"

5.  -2.2304425   0.  -0.1585127
```

5. b. Program to perform KL transform for the given 2D matrix

Code

```
clc;
f=[1,1,1,1;1,1,1,1;1,1,1,1;1,1,1,1];
N=4
for k=0:N-1
   for l=0:N-1
      w(k+1,l+1)=int(cos((2*%pi*k*1)/N)-%i*sin((2*%pi*k*1)/N))
   end
end

disp(w,'Kernel Matrix of 2d DFT=')
F=w*f*w
disp(F,'2D DFT of given 2D image=')
```

Output

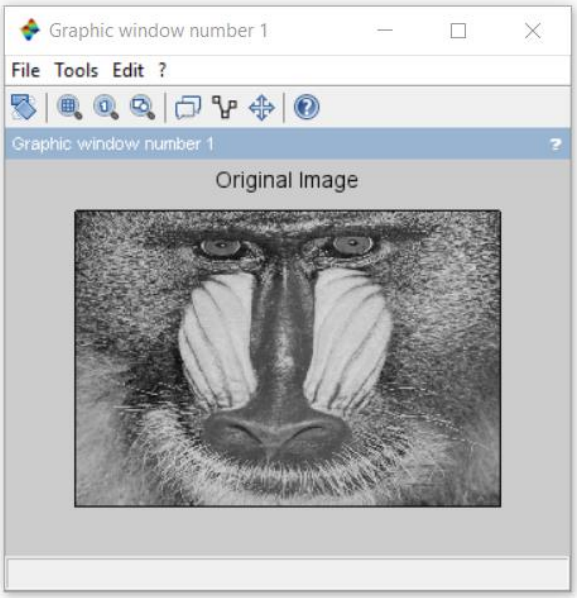
```
  "Kernel Matrix of 2d DFT="

  1. + 0.i   1. + 0.i   1. + 0.i   1. + 0.i

  0. - i     0. - i     0. - i     0. - i

 -1. + 0.i  -1. + 0.i  -1. + 0.i  -1. + 0.i

  0. + i     0. + i     0. + i     0. + i

"2D DFT of given 2D image="

  0. + 0.i   0. + 0.i   0. + 0.i   0. + 0.i

  0. + 0.i   0. + 0.i   0. + 0.i   0. + 0.i

  0. + 0.i   0. + 0.i   0. + 0.i   0. + 0.i

  0. + 0.i   0. + 0.i   0. + 0.i   0. + 0.i
```

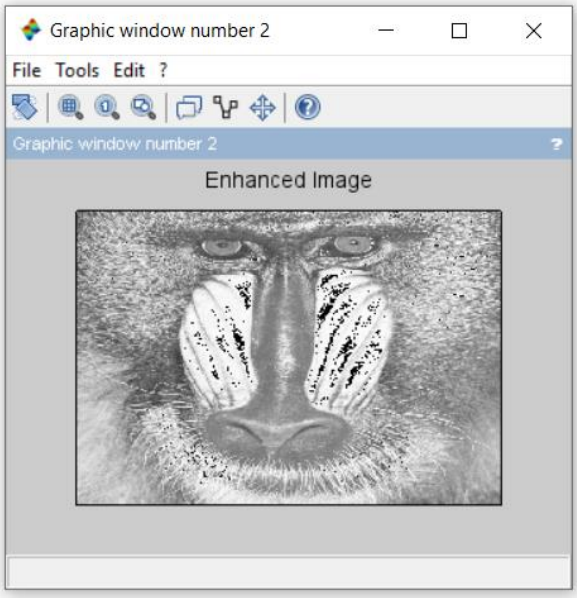
6. a. Brightness enhancement of an image

Code

```
clc;
close;
a=imread((getIPCVpath() +"/images/"+'baboon.png'))
a=rgb2gray(a);
b=double(a)+50;
b=uint8(b);
figure(1)
imshow(a);
title('Original Image')
figure(2)
imshow(b);
title('Enhanced Image')
```

Output



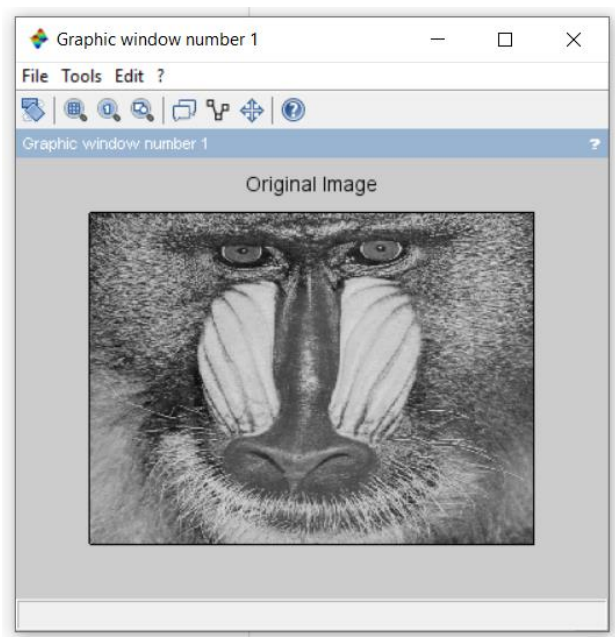


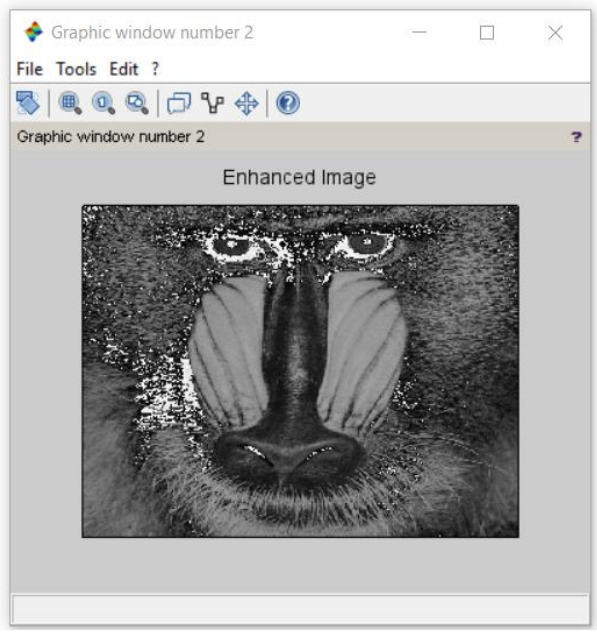
6. b. Brightness suppression of an image

Code

```
clc;
close;
a=imread((getIPCVpath() +"/images/"+'baboon.png'))
a=rgb2gray(a);
b=double(a)-50;
b=uint8(b);
figure(1)
imshow(a);
title('Original Image')
figure(2)
imshow(b);
title('Enhanced Image')
```

Output



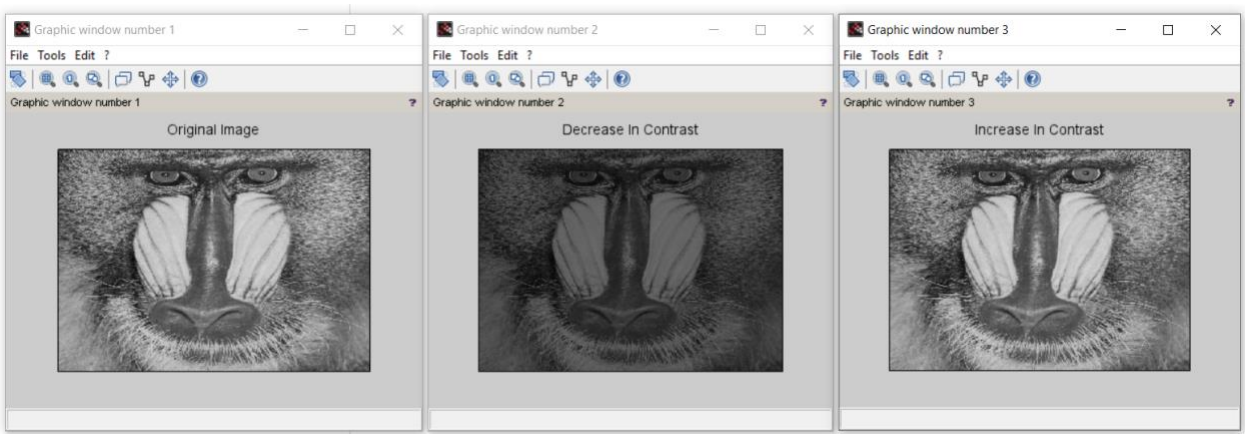


6. c. Contrast Manipulation

Code

```
clc;
close;
a=imread((getIPCVpath() +"/images/"+'baboon.png'))
a=rgb2gray(a);
b=double(a)*0.5;
b=uint8(b);
c=double(b)*2;
c=uint8(c);
figure(1)
imshow(a);
title('Original Image')
figure(2)
imshow(b);
title('Decrease In Contrast')
figure(3)
imshow(c);
title('Increase In Contrast')
```

Output

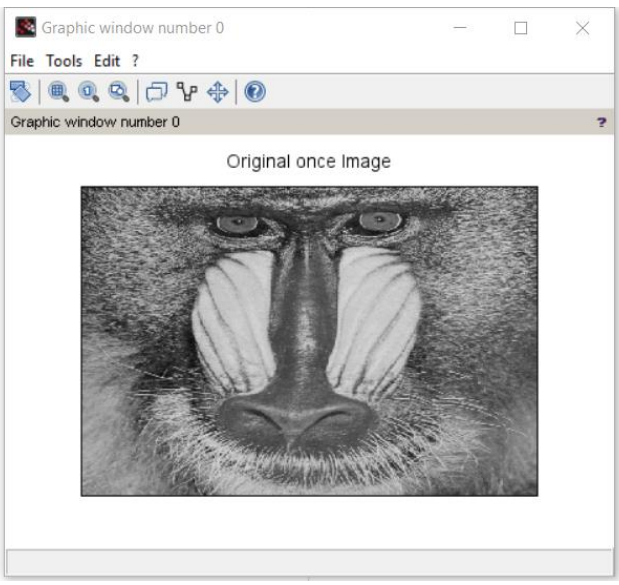


6. d. Image negative

Code

```
clc;
close;
a=imread(fullpath(getIPCVpath() +"/images/"+'baboon.png'));
a=rgb2gray(a);
k=255-double(a);
k=uint8(k);
imshow(a);
title('Original once Image')
figure(2)
imshow(k);
title('Negative Of Original Image')
```

Output



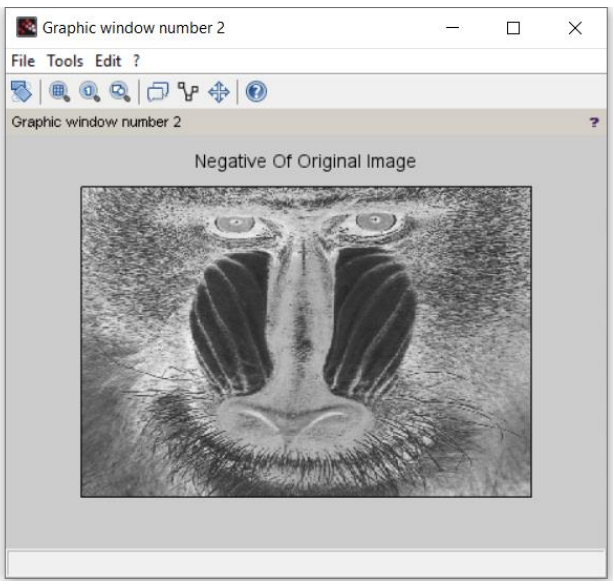
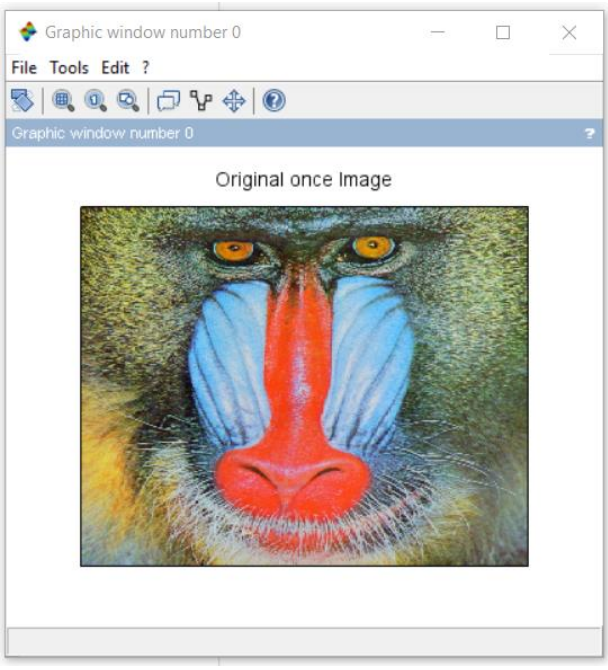


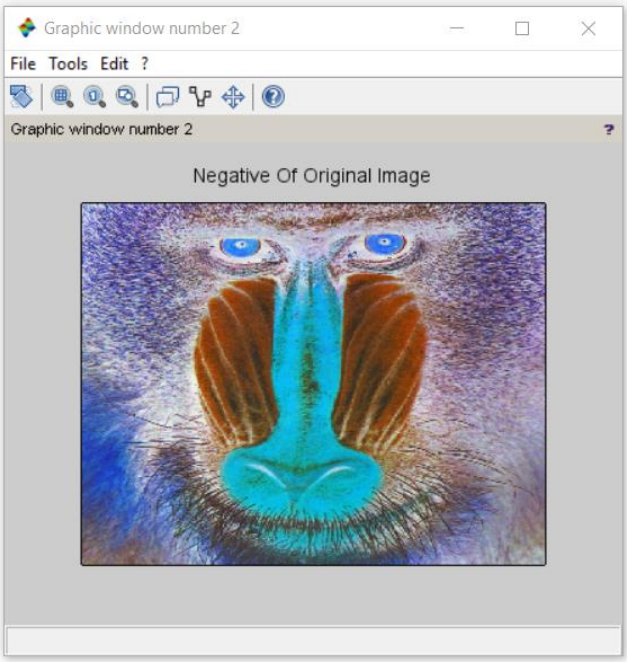
Image Negative of color image

Code

```
clc;
close;
a=imread(fullpath(getIPCVpath() +"/images/"+'baboon.png'));
k=255-double(a);
k=uint8(k);
imshow(a);
title('Original once Image')
figure(2)
imshow(k);
title('Negative Of Original Image')
```

Output



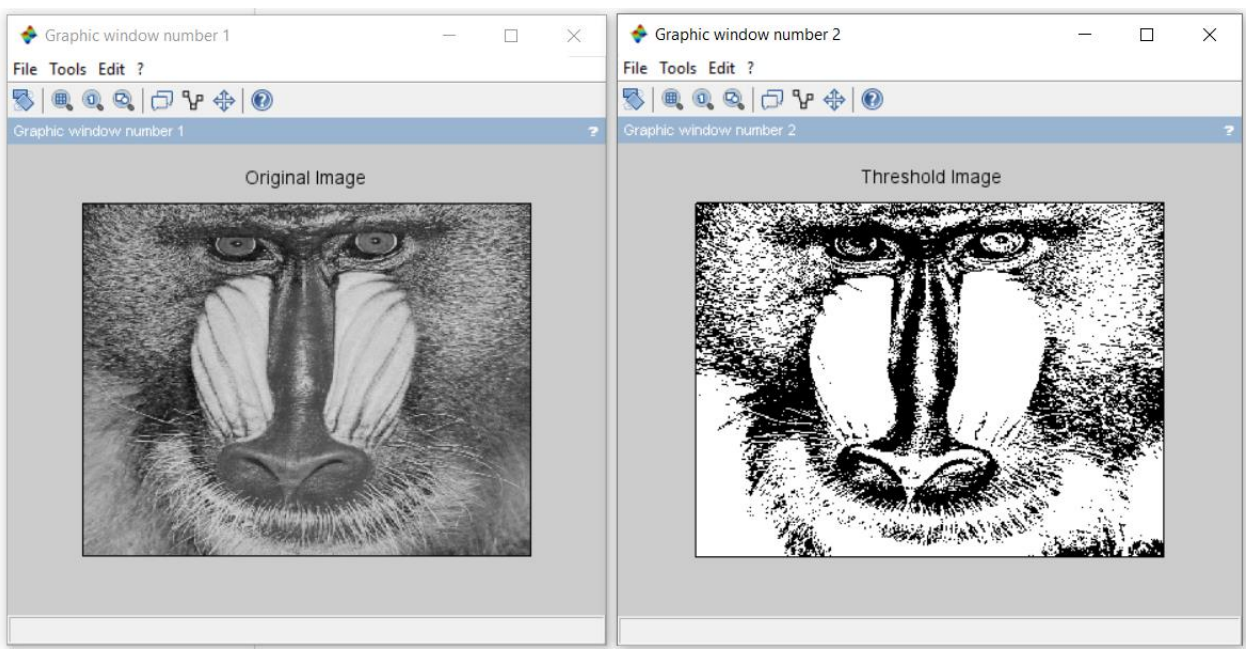


7. a. Perform threshold operation

Code

```
clc;
close;
a=imread((getIPCVpath() +"/images/"+'baboon.png'))
a=rgb2gray(a);
[ m n ] = size (a);
for i =1 :m
   for j = 1 : n
      if(a(i,j)< 100)
         b(i,j)=0;
      else
         b(i,j)=255;
      end
    end
end
figure(1)
imshow(a);
title('Original Image');
figure(2)
imshow(b);
title('Threshold Image');
```

Output

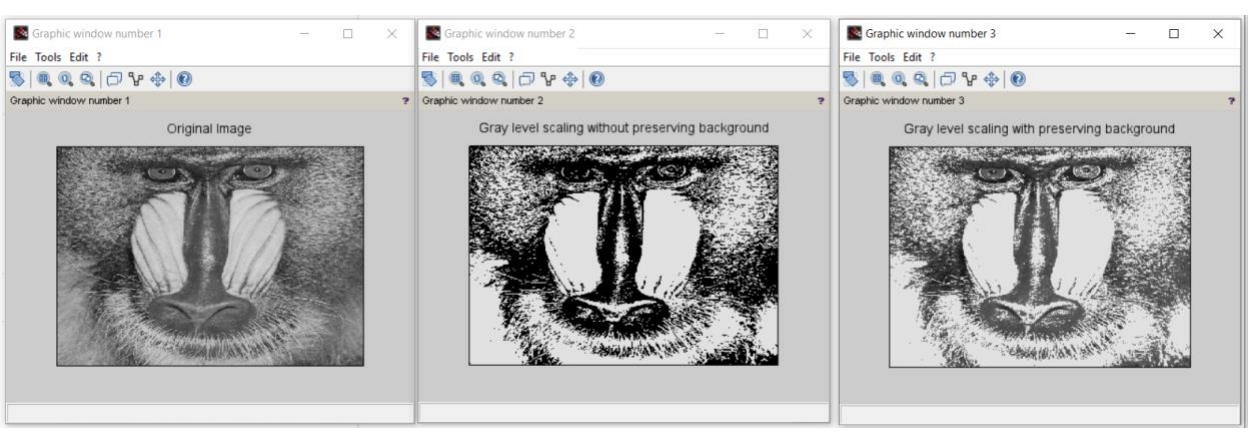


7. b. perform gray level slicing without background

Code

```
clc;
x=imread(fullpath(getIPCVpath() +"/images/"+'baboon.png'));
x=rgb2gray(x);
y=double(x);
[m,n] =size(y);
L=max(max(x));
a=round(L/2);
b=L;
for i = 1:m
for j = 1:n
if(y(i,j)>= a & y(i,j)<=b )
z (i,j)=L;
else
z(i,j)=0;
end
end
end
z=uint8(z);
for i = 1:m
for j = 1:n
if(y(i,j)>= a & y(i,j)<=b )
w (i,j)=L;
else
w(i,j)=y(i,j);
end
end
end
z=uint8(z);
w=uint8(w);
figure(1)
imshow(x);
title('Original Image')
figure(2)
imshow(z);
title(' Gray level scaling without preserving background')
figure(3)
imshow(w);
title(' Gray level scaling with preserving background')
```

Output



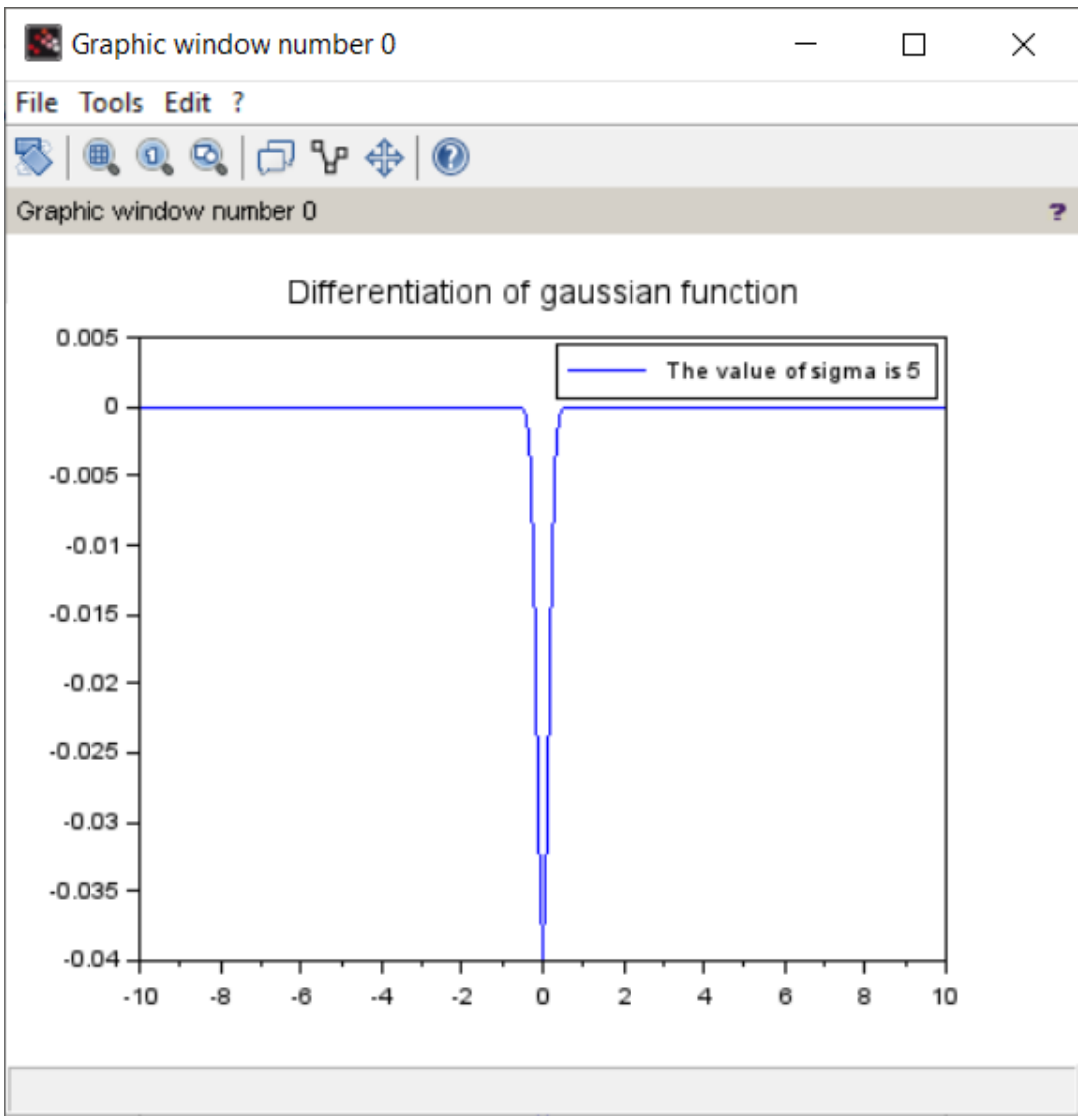
8. Image Segmentation

8. a. Differentiation of Gaussian function

Code

```
clc;
close;
sigma=input('Enter the value of sigma:');
i=-10:.1:10;
j=-10:.1:10;
r=sqrt(i.*i+j.*j);
y=(1/(sigma^2))*(((r.*r)/sigma^2)-1).*exp(-r.*r/2*sigma^2);
plot(i, y)
legend(sprintf('The value of sigma is %g',sigma))
xtitle(' Differentiation of gaussian function ')
```

Output



8. b. Shape Of DOG Filter

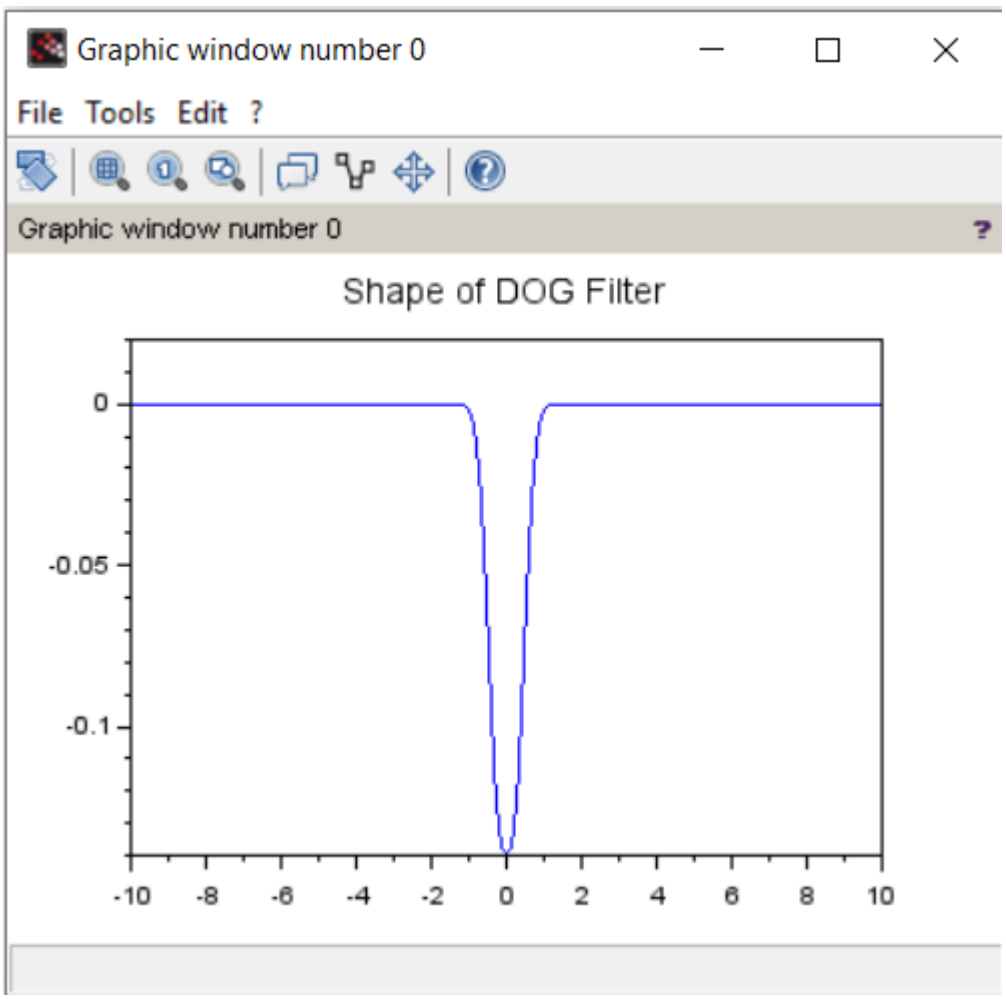
Code

```
clc;
close;
sigma1 =input(' Enter the value of sigma1 : ')
sigma2 =input(' Enter the value of sigma2 : ')
i= -10:.1:10;
j= -10:.1:10;
r=sqrt(i.*i+j.*j);
y1 = (1/( sigma1 ^2))*(((r.*r)/sigma1 ^2) -1).*exp(-r.*r/2* sigma1 ^2);
y2 = (1/( sigma2 ^2))*(((r.*r)/sigma2 ^2) -1).*exp(-r.*r/2* sigma2 ^2);
y = y1 -y2;
plot(i,y)
xtitle(' Shape of DOG Filter ')
```

Output

Enter the value of sigma1 : 2

Enter the value of sigma2 : 3



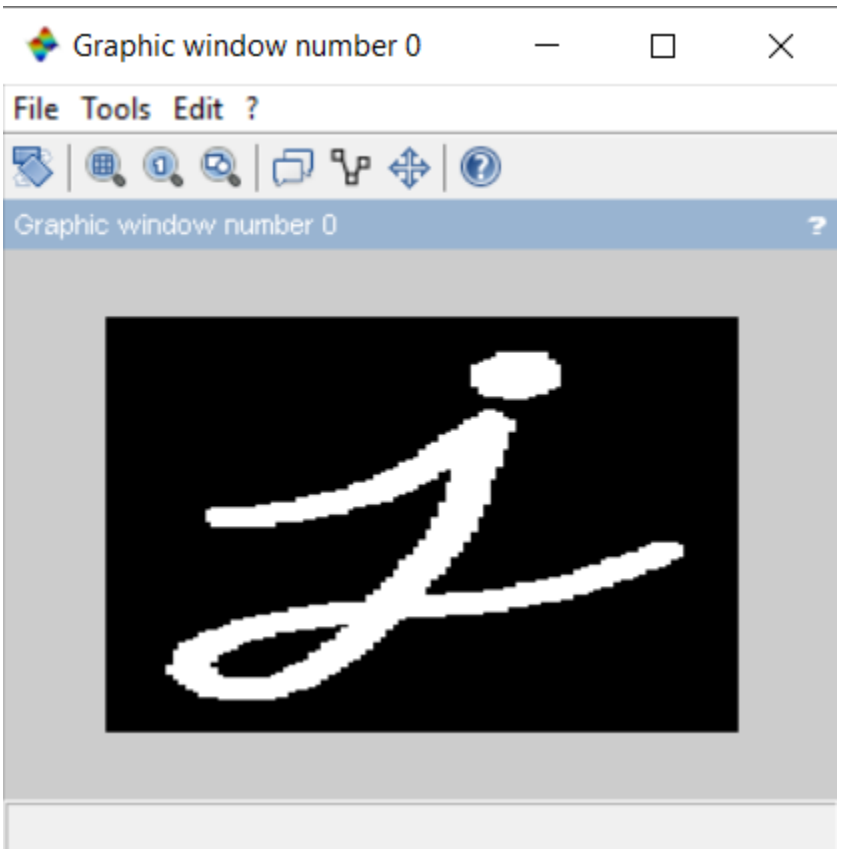
9. Binary Image Processing

9. a. Image Dilation

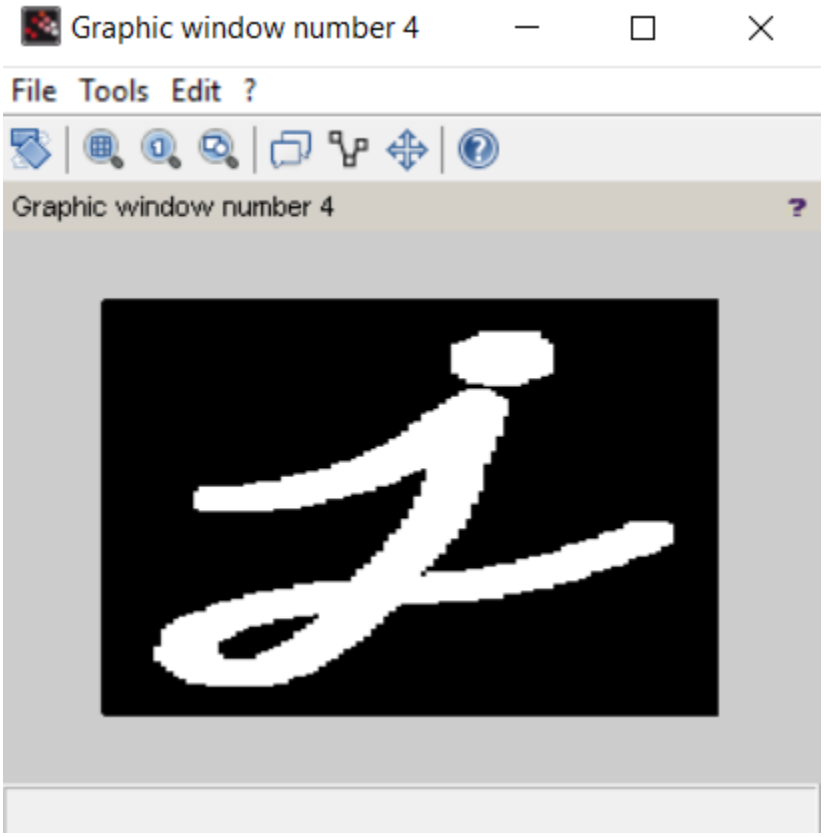
Code

```
S=imread(fullpath(getIPCVpath() +"/images/"+'morpex.png'))
figure(0);
imshow(S);
se = imcreatese('ellipse',9,9);
c= imdilate(S,se);
figure(1);
imshow(c);
```

Output Original Image



Dilated Image

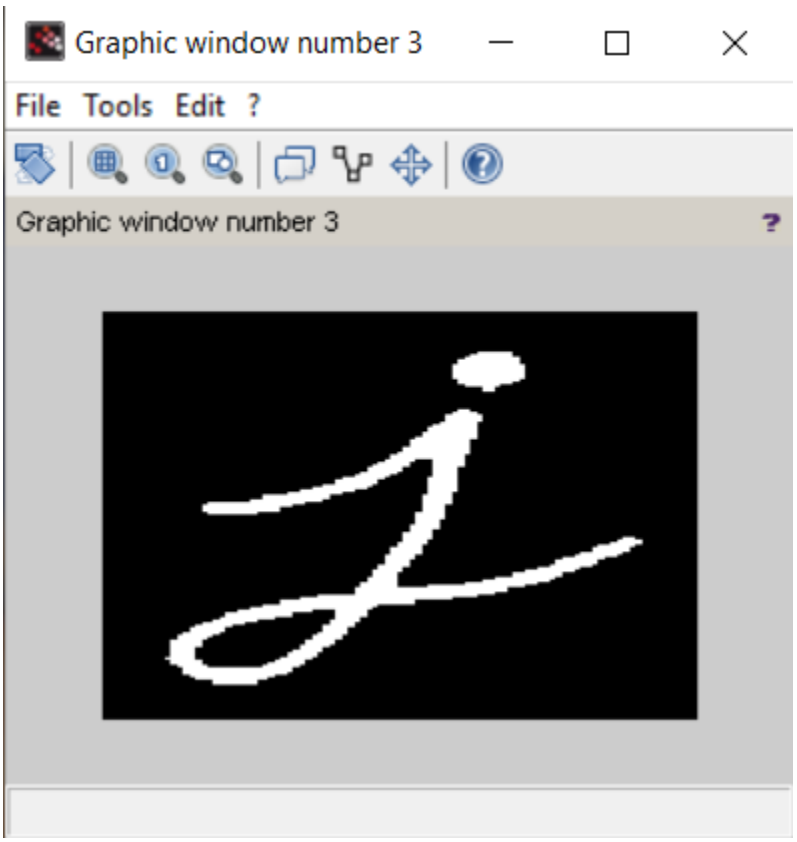


9. b. Image Erosion

Code

```
S=imread(fullpath(getIPCVpath() +"/images/"+'morpex.png'))
figure(0);
imshow(S);
se= imcreatese('rect',3,3);
b = imerode(S,se);
figure(1);
imshow(b);
```

Output Eroded Image

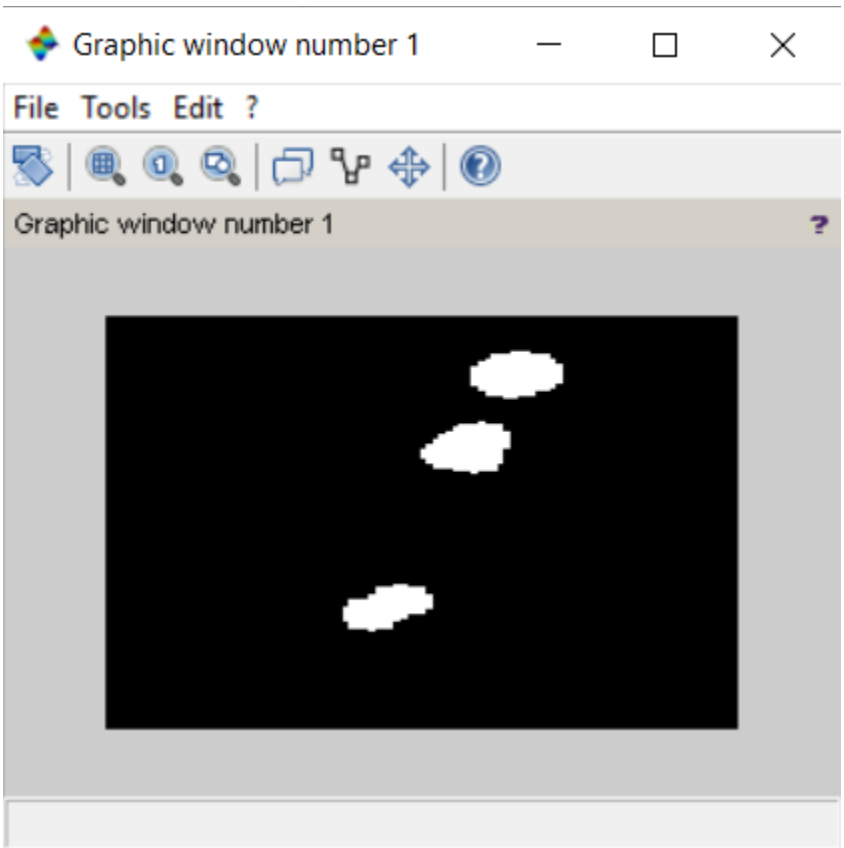


9. c. Image Opening

Code

```
S=imread(fullpath(getIPCVpath() +"/images/"+'morpex.png'))
figure(0);
imshow(S);
se = imcreatese('ellipse',9,9);
S2 = imopen(S,se);
figure(1);
imshow(S2);
```

Output

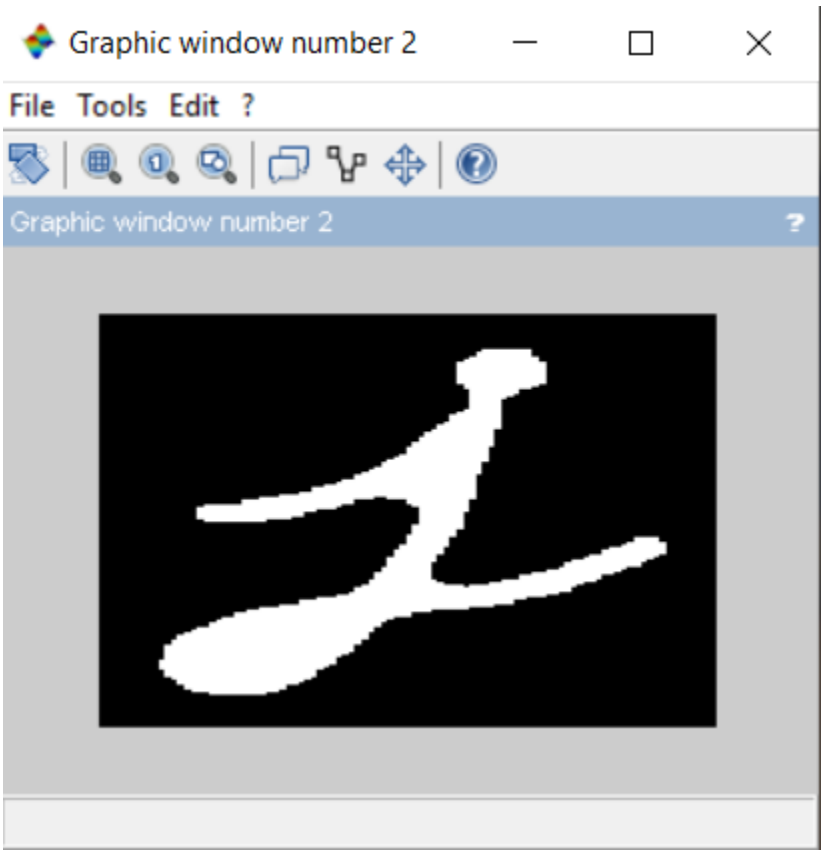


9. d. Image Closing

Code

```
S=imread(fullpath(getIPCVpath() +"/images/"+'morpex.png'))
figure(0);
imshow(S);
se= imcreatese('ellipse',11,11);
S3 = imclose(S,se);
figure(2);
imshow(S3);
```

Output



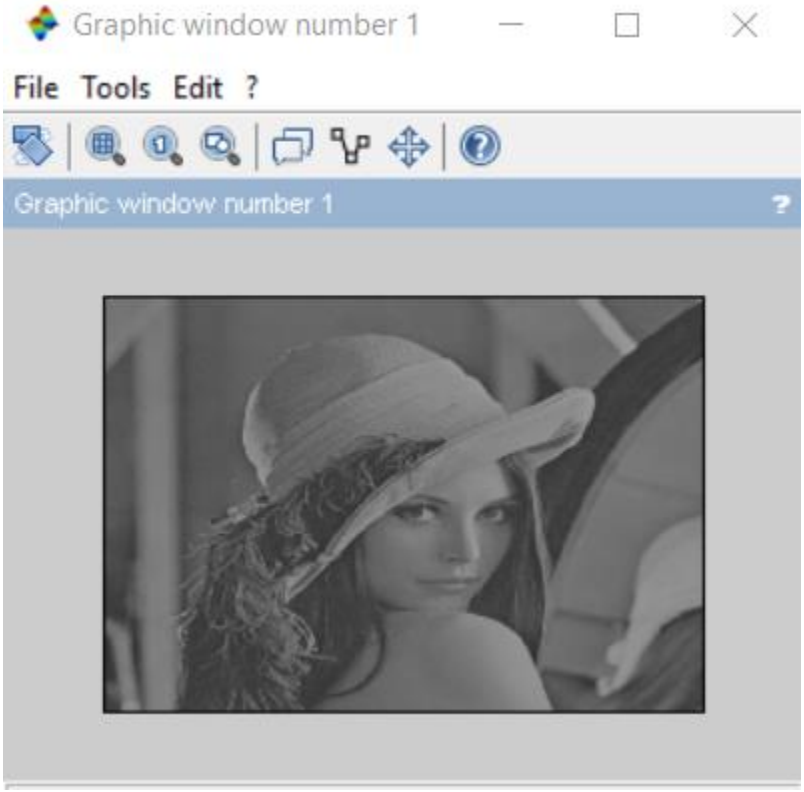
10. Color Image processing

10. a. Histogram of Gray Scale Image

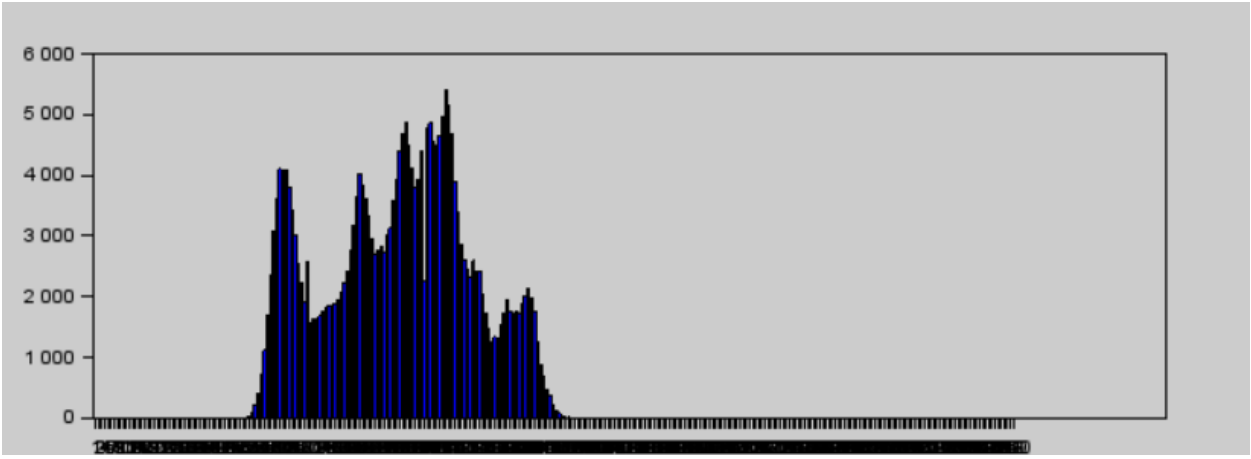
Code

```
a=imread(fullpath(getIPCVpath() +"/images/"+'Lena_dark.png'));
a=double(a);
big=max(max(a));
[row col] = size(a);
c=row*col;
h=zeros(1,300);
z=zeros(1,300);
for n= 1:1:row
for m= 1:1:col
if a (n,m)==0 then
   a(n,m)=1;
end
end
end
for n=1:1:row
   for m=1:1:col
      t=a(n,m);
      h(t)=h(t)+1;
end
end
figure(1);
imshow(uint8(a));
figure(2);
bar(h);
```

Original Grayscale Image



Original Grayscale Image Histogram

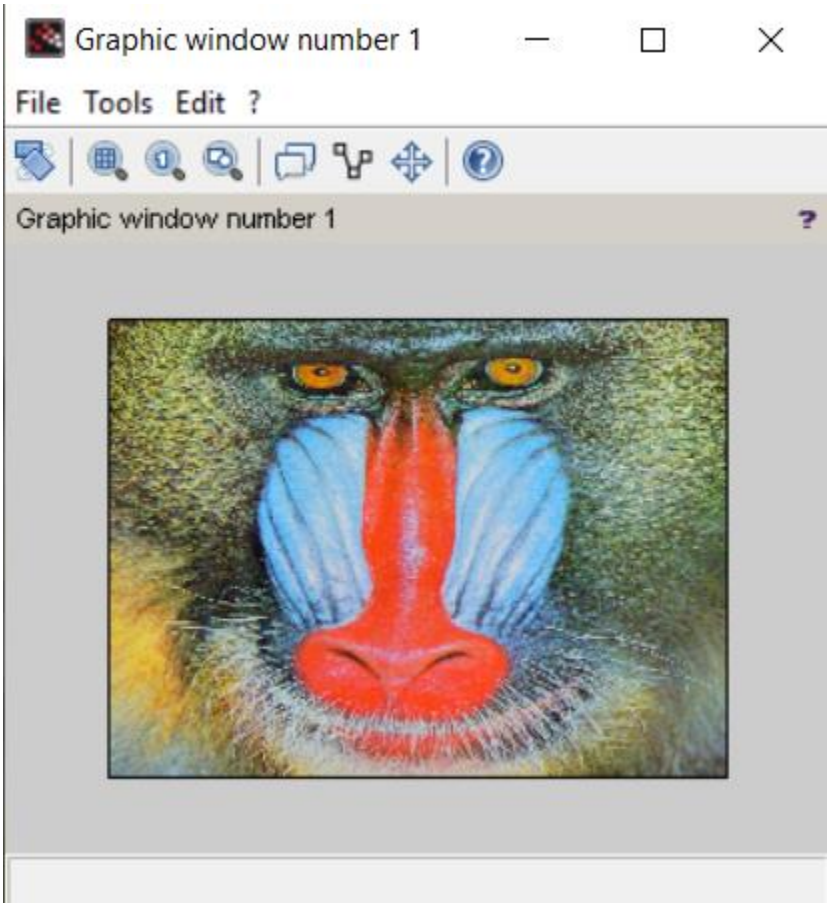


10. b. Histogram of Color Image

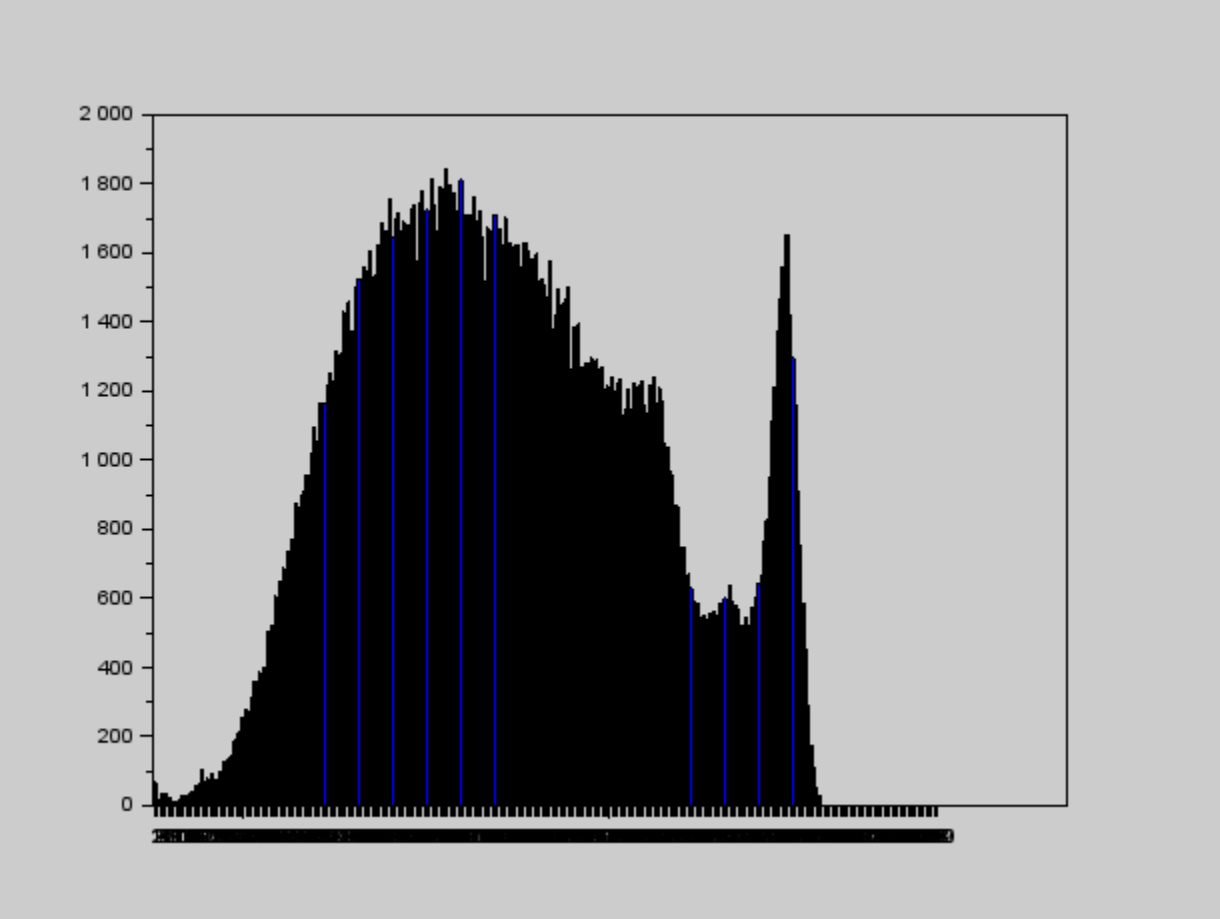
Code

```
a=imread(fullpath(getIPCVpath() +"/images/"+'baboon.png'));
a=double(a);
big=max(max(a));
[row col] = size(a);
c=row*col;
h=zeros(1,300);
z=zeros(1,300);
for n= 1:1:row
for m= 1:1:col
if a (n,m)==0 then
   a(n,m)=1;
end
end
end
for n=1:1:row
   for m=1:1:col
      t=a(n,m);
      h(t)=h(t)+1;
end
end
figure(1);
imshow(uint8(a));
figure(2);
bar(h);
```

Original Color Image



Color Image Histogram

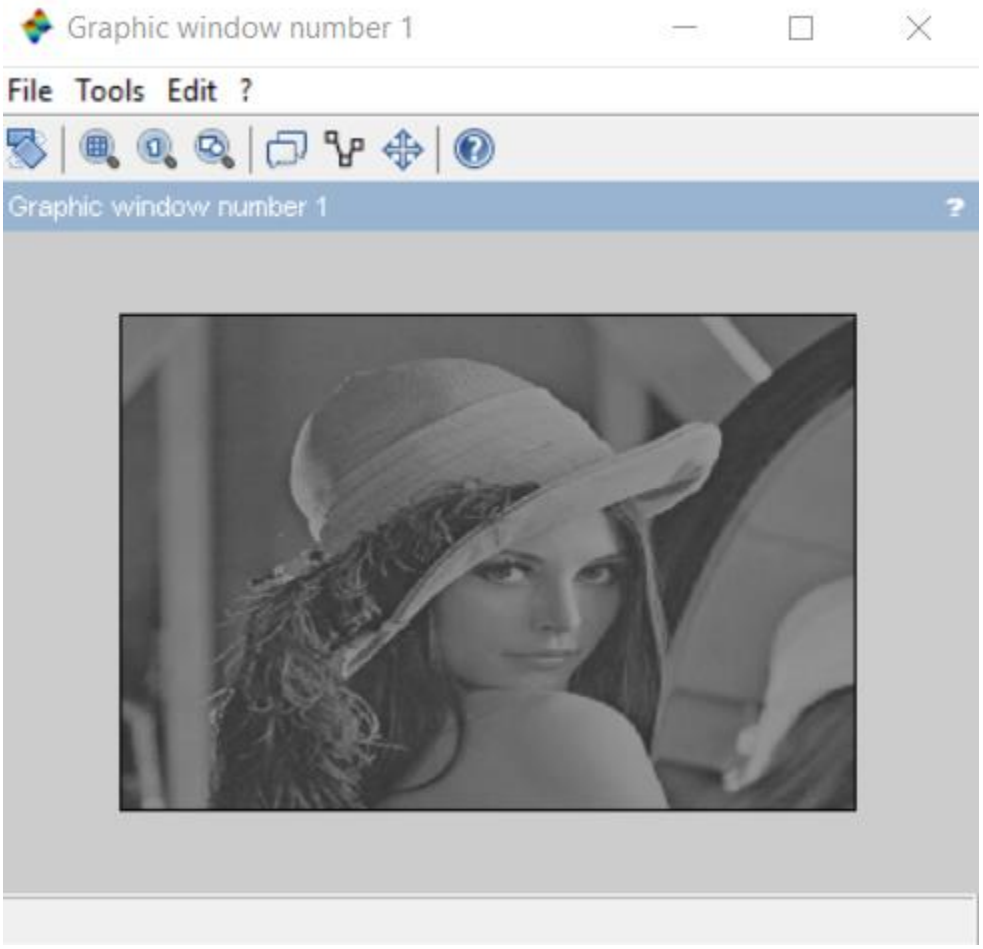


10.c Histogram Equalization with imhistequal function

Code

```
S=imread(fullpath(getIPCVpath() +"/images/"+'Lena_dark.png'));
J = imhistequal(S);
figure(1);
imshow(S);
figure(2);
imshow(J);
```

Original Image (Lena_dark)



Original Image Histogram (Lena_dark)

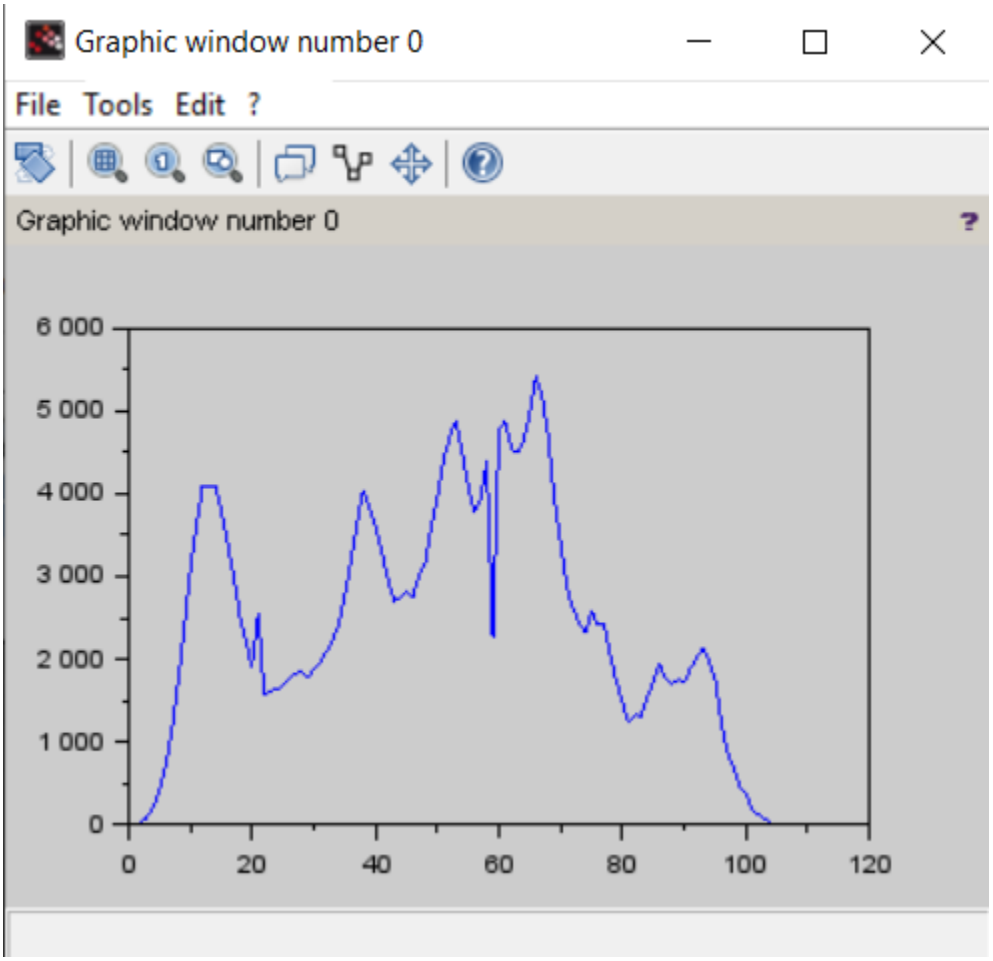
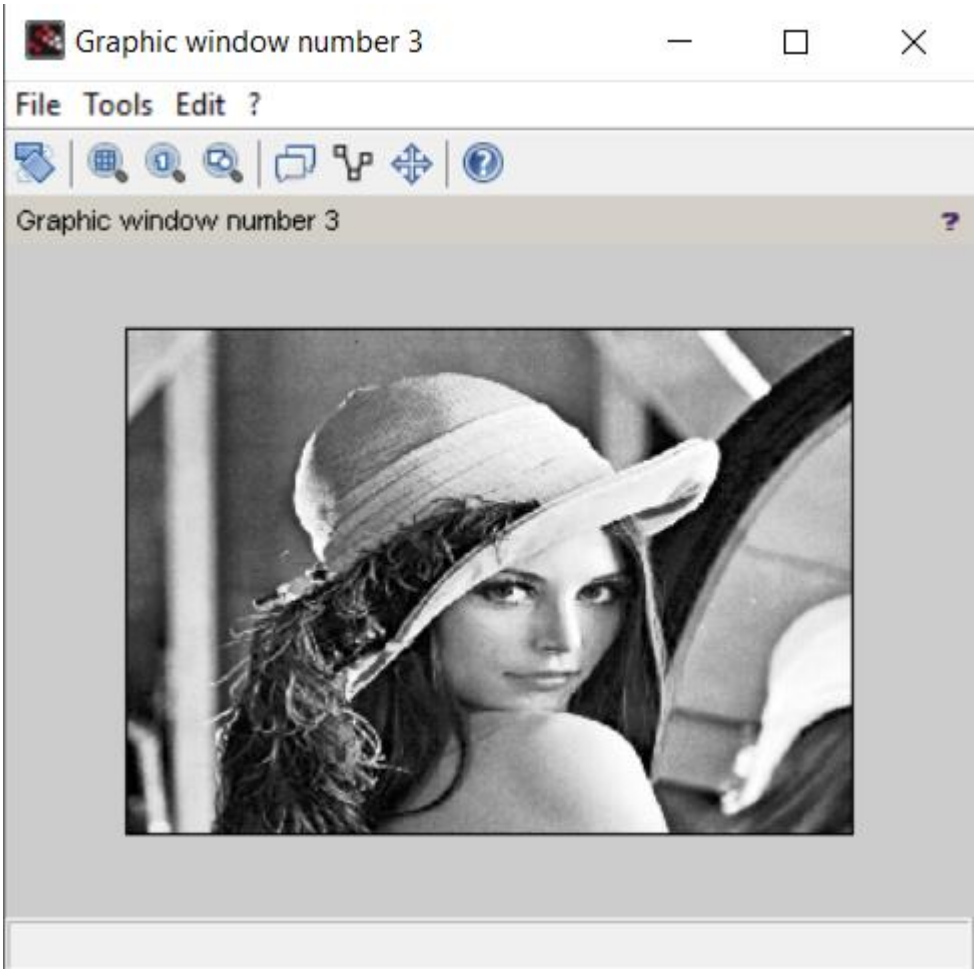
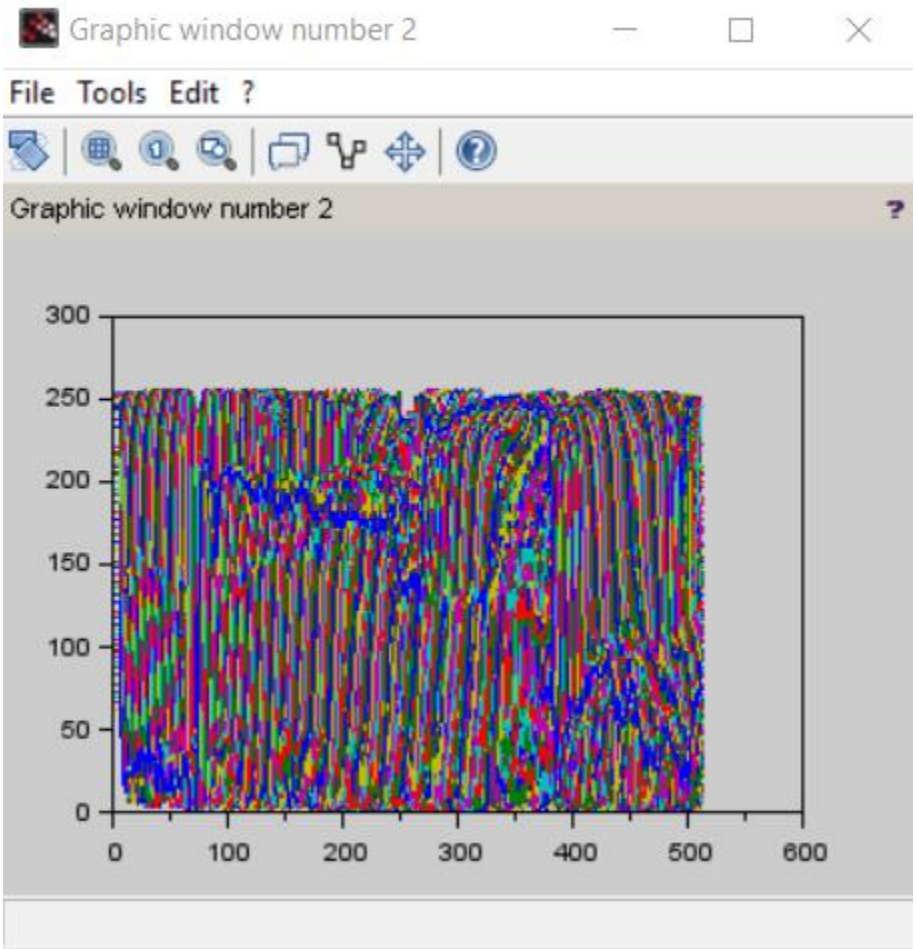


Image Enhancement After Procedure Of Histogram Equalization (Lena_dark)



Histogram After Procedure Of Histogram Equalization (Lena_dark)



10.d Histogram Equalization Without Inbuilt Function

Equalization is a process that attempts to spread out the gray levels in an image so that they are evenly distributed across their range. Histogram Equalization reassigns the brightness values of pixels based on the image histogram. Histogram Equalization is a technique where the histogram of the resultant image is as flat as possible. Histogram Equalization provides more visually pleasing results across a wider range of images.

Procedure to Perform Histogram Equalization Histogram:

Equalization is done by performing the following steps:

1. Find the running sum of the histogram values.

2. Normalize the values from Step (1) by dividing by the total number of pixels.

3. Multiply the values from Step (2) by the maximum gray-level value and round.

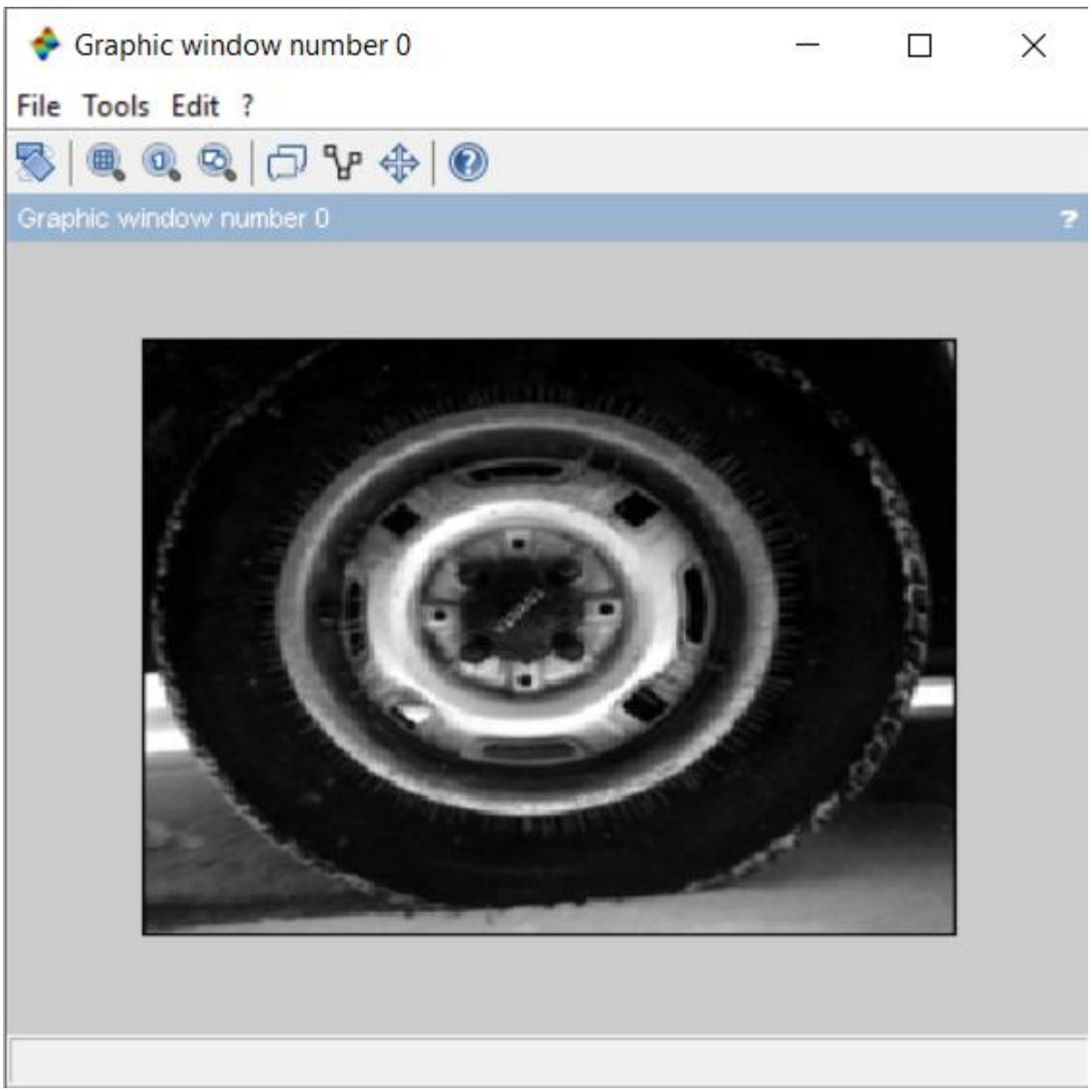
4. Map the gray level values to the results from Step (3) using a one-to-one correspondence

Scilab Code:

```
a=imread('tire.tif'); //read image
figure(0);
imshow(a); // display image
histogram =histc(a);
plot(histogram);
a=double(a); //converts inttype integers or booleans into decimal encoding

big=max(max(a)); // maximum
[r c] = size(a); //matrix size
tot = r*c  //multiply rows and columns
h=zeros(1,256); //store histogram values
z=zeros(1,256);
for i=1:1:r
   for j=1:1:c
      if a(i,j)==0
         a(i,j)=1;
      end
   end
end
for i=1:1:r
   for j=1:1:c
      t=a(i,j);
      h(t)=h(t)+1;
      end
end
pdf=h/tot; //probability distribution function
cdf(1)=pdf(1); //cumulative distribution function i.e.,running sum

for i=2:1:big
```

```
    cdf(i)=pdf(i)+cdf(i-1);
end
new=round(cdf*big); //rounds the elements of x to the nearest integers.
new=new+1;
for i=1:1:r
    for j=1:1:c
        temp=a(i,j);
        b(i,j)=new(temp);
        t=b(i,j);
        z(t)=z(t)+1;
    end
end
subplot(2,1,2);
plot(z);
figure(1);
imshow(b);
```

Original Image (Tire)



Original Image Histogram (Tire)

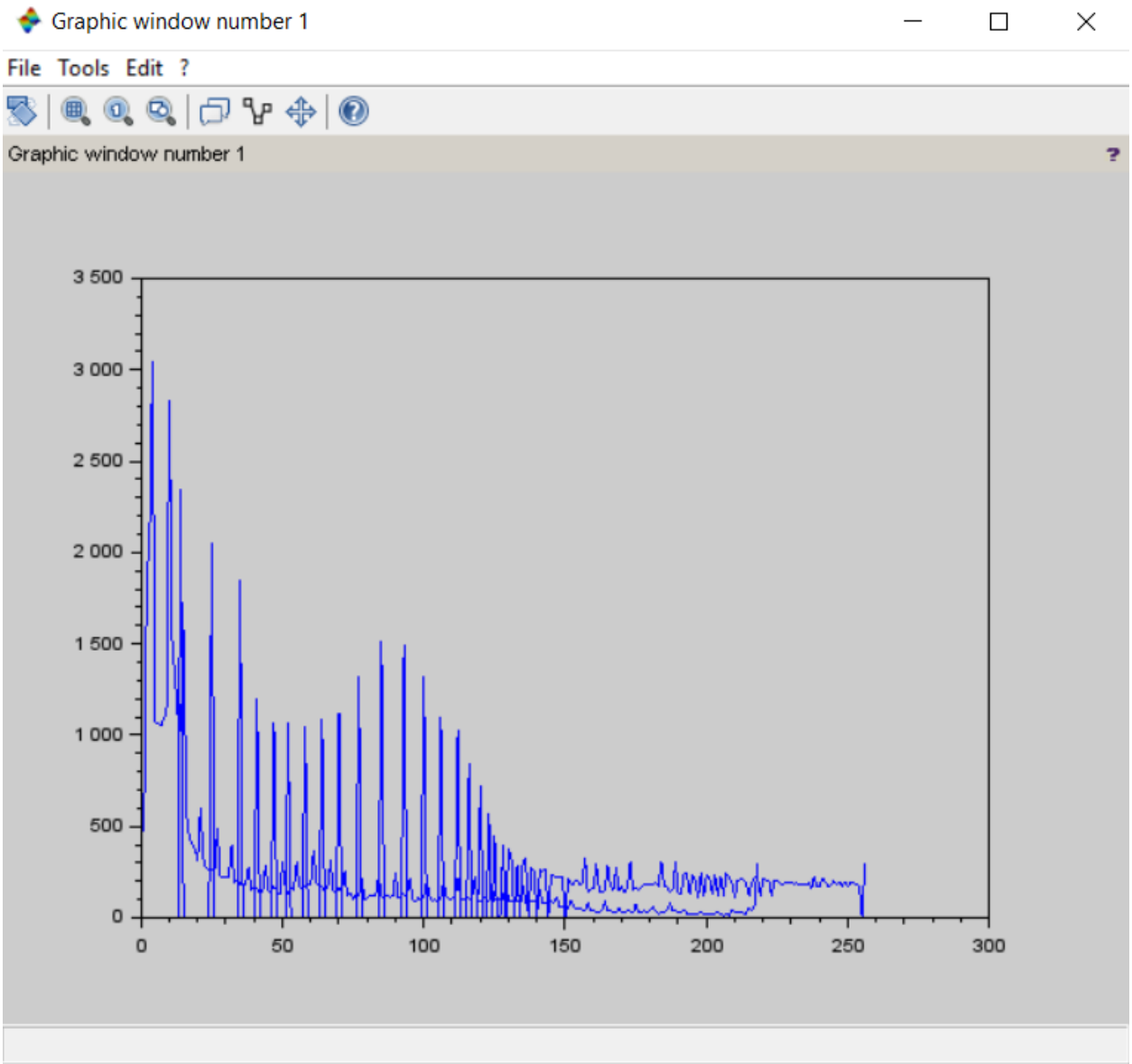
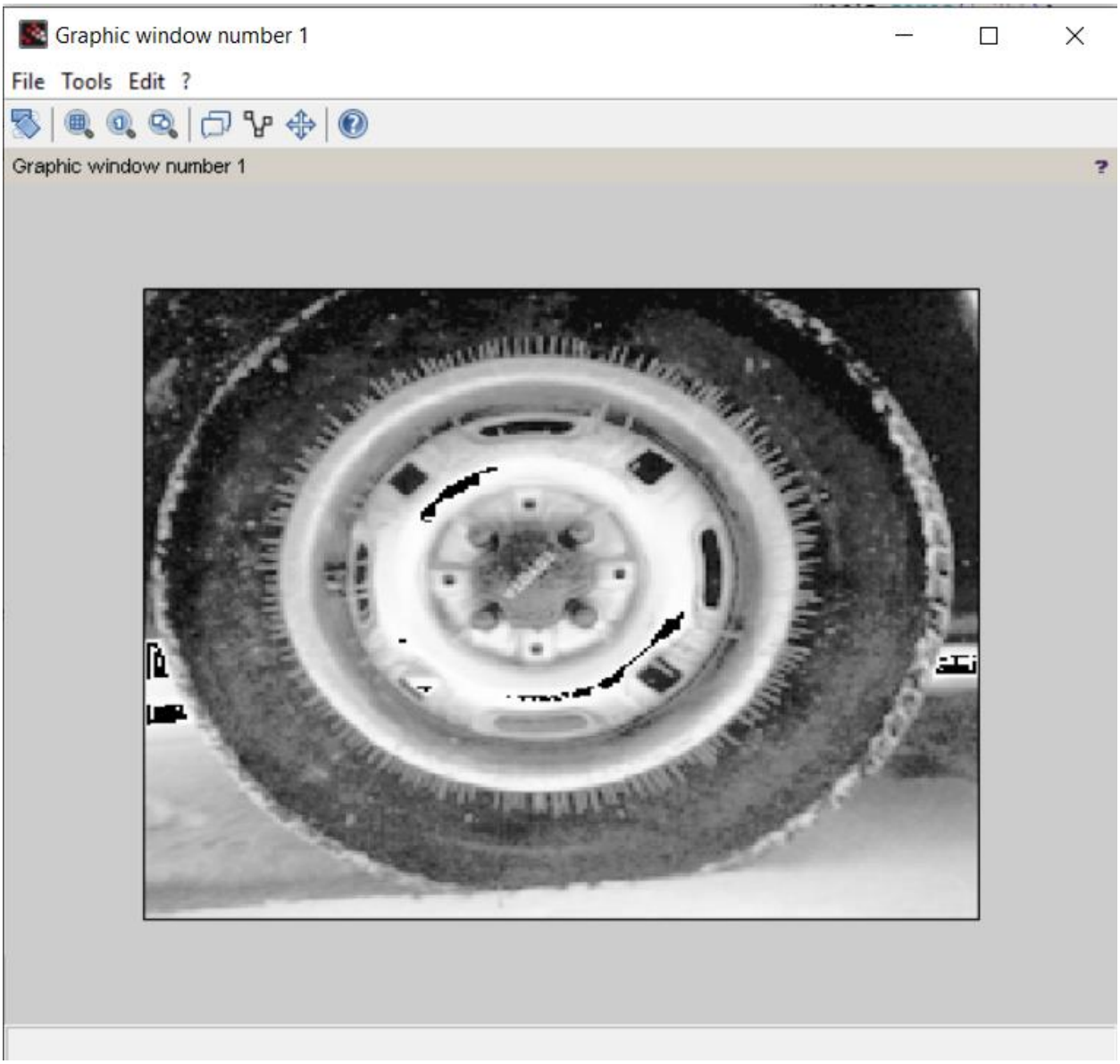
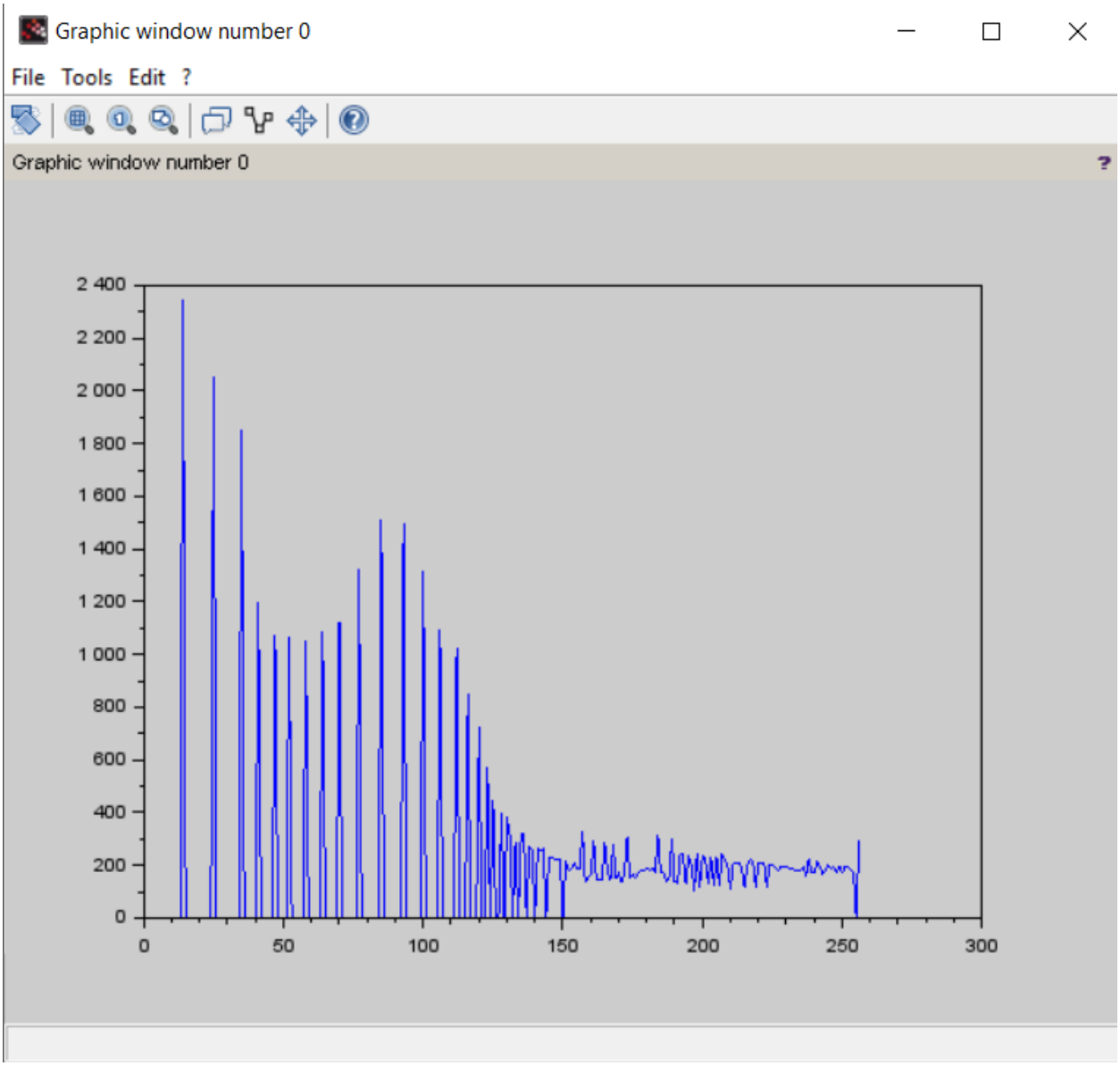


Image Enhancement After Procedure Of Histogram Equalization (Tire)



Histogram After Procedure Of Histogram Equalization (Tire)

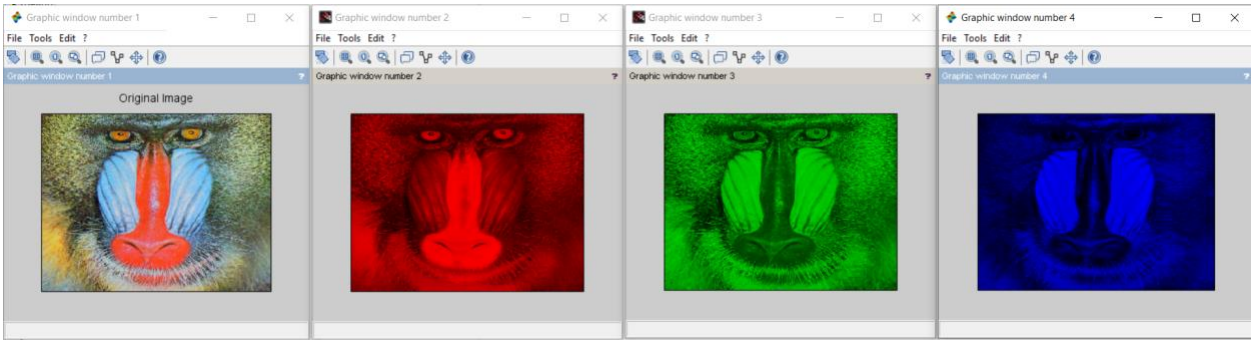


10. e Read an RGB image and Extract the three color components red green blue

Code

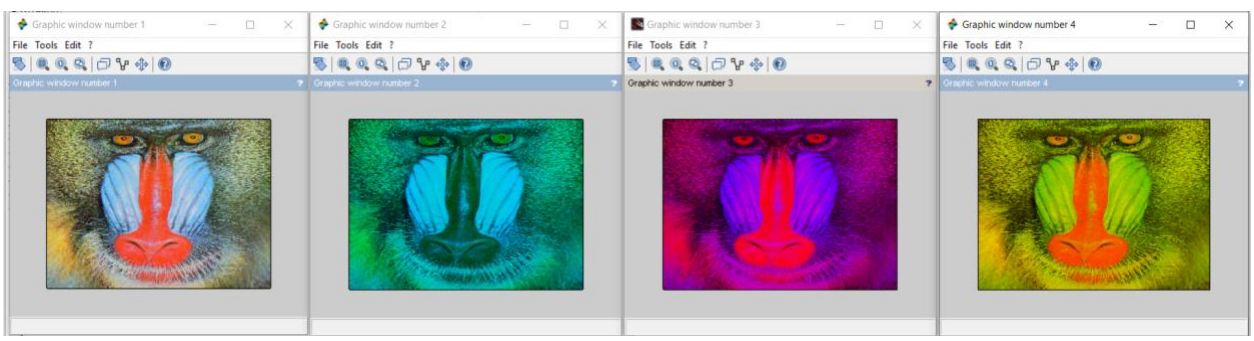
```
clc;
close;
RGB=imread((getIPCVpath() +"/images/"+'baboon.png'));
R=RGB;
G=RGB;
B=RGB;
R(:,:,2)=0;
R(:,:,3)=0;
G(:,:,1)=0;
G(:,:,3)=0;
B(:,:,1)=0;
B(:,:,2)=0;
figure(1)
imshow(RGB);
title('Original Image')
figure(2)
imshow(R);
figure(3)
imshow(G);
figure(4)
imshow(B);
```

Output



10. f Read a Color image and separate the color image into red green and blue planes

Code

```
clc;
close;
RGB=imread((getIPCVpath() +"/images/"+'baboon.png'));
a1=RGB;
b1=RGB;
c1=RGB;
a1(:,:,1)=0;
b1(:,:,2)=0;
c1(:,:,3)=0;
figure(1)
imshow(RGB);
figure(2)
imshow(a1);
figure(3)
imshow(b1);
figure(4)
imshow(c1);
```

Output